

Avinash Kumar Singh

Computer Science Engineering
Undergraduate

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Summary

Computer Science Engineering undergraduate with a strong grasp of object-oriented programming, data structures, and design patterns, and experience building production full-stack applications across Java, Python, and TypeScript/JavaScript (React, Next.js, Node.js, Express, Flask, REST APIs, SQL/PostgreSQL). Also skilled in cybersecurity, having run 30+ network and web application security assessments identifying 150+ findings using Nmap, Wireshark, Burp Suite, and OWASP ZAP under OWASP Top 10 methodology, with CVSS-based vulnerability reporting. Experienced writing and reviewing pull requests, applying test-driven development, and communicating clearly with engineers and stakeholders. Seeking a Software Engineer Internship to ship measurable-impact production software.

Education

2023–2027 **B.Tech in Computer Science Engineering (Cyber Security and Digital Forensics)**, VIT Bhopal University, Kothri Kalan, CGPA: 8.20

Technical Skills

Programming Languages	Java, C++, Python, JavaScript, TypeScript, SQL, LaTeX
Web & Databases	HTML, CSS, React.js, Next.js, Node.js, Express.js, Flask, Prisma, REST API, HTTP, MySQL, SQLite, PostgreSQL, MongoDB
Software Engineering	Object-Oriented Programming (OOP), Design Patterns, Test-Driven Development (TDD), Microservices, Git, GitHub, pull requests, code review, unit testing, test coverage, Linux, Agile collaboration, Systems design, Data structures and algorithms, CI/CD, Docker
Security	Kali Linux, Nmap, Wireshark, Burp Suite, OWASP ZAP, Metasploit, OWASP Top 10, CVSS, JWT

Experience

Apr 2026 – Present **Cybersecurity Analyst Intern**, Zorvyn FinTech Pvt. Ltd., Remote

- Conducted 30+ network and web application security assessments across 20+ endpoints using Nmap, Wireshark, Burp Suite, and OWASP ZAP, identifying 150+ security findings aligned with the OWASP Top 10.
- Analyzed 500,000+ network packets and performed reconnaissance on 20+ hosts, uncovering 25+ network misconfigurations, exposed services, and anomalous traffic patterns.
- Executed 50+ web application penetration testing scenarios, validating vulnerabilities including SQL Injection, Cross-Site Scripting (XSS), Broken Access Control, and Security Misconfiguration.
- Authored 30+ technical vulnerability reports with CVSS-based risk scoring, proof-of-concept evidence, and prioritized remediation guidance, reducing average remediation turnaround by 30%.
- Collaborated with 5+ developers and security team members in Linux environments to verify remediation, retest vulnerabilities, and improve the organization's overall security posture.

Jan 2026 – **Frontend Developer & Security Lead — Academic Research Project (Breast Cancer Classification System)**, *VIT Bhopal University*

- Developed a React.js + Flask web application for breast cancer classification with RESTful API integration and an interactive prediction dashboard, connecting to a backend Random Forest classifier (96.49% test accuracy) trained on the 569-sample UCI Wisconsin Breast Cancer dataset.
- Designed secure input validation (required fields, numeric constraints, range checks, backend validation) to ensure data integrity and reliable ML inference, alongside authentication and secure data handling to guard against common web vulnerabilities.
- Improved application efficiency by implementing single-instance model loading with Joblib, reducing API response time by 90%.
- Performed frontend testing, fixed bugs, and optimized performance across the interactive prediction dashboard for a smooth end-to-end user experience.

Projects

Oct 2025 – **CuraBreast AI — AI-Powered Women's Healthcare Assistant**

- Interviewed 139+ women, primarily in rural areas, uncovering that low health literacy and social stigma made women reluctant to discuss breast health with family, often delaying care entirely.
- Reframed the product as a private, judgment-free "first step," letting women self-assess symptoms and understand risk before ever needing to raise it with family or a doctor.
- Built a full-stack web app (Node.js, Express, React, PostgreSQL) with a weighted, multi-factor risk-scoring engine spanning 9 clinical, genetic, and lifestyle indicators to classify users into low/moderate/high risk tiers.
- Secured the platform with JWT authentication, bcrypt password hashing, rate limiting, and XSS/Helmet protections, and built an admin panel for user and assessment oversight.
- Delivered a 10-hospital locator with map directions, appointment booking, and filterable assessment reports with CSV export, generating personalized recommendations for 50+ test users.
- github.com/avinash09-cmd/curabreast-ai

Feb 2025 – **N.E.S.T. (Neighbourhood Emergency and Safety Tool)**

- Spoke with residents across 17 housing societies and found that reporting issues through third-party channels was slow, opaque, and time-consuming, with no visibility into resolution status.
- Built a full-stack civic platform (Next.js 15, TypeScript, Prisma, PostgreSQL) with NextAuth-based role access for Residents, Admins, and Super Admins, supporting 100+ simulated complaint records.
- Designed a 3-step async AI pipeline (Claude Sonnet) that auto-classifies incoming complaints by category, runs sentiment analysis, and generates a 0–100 risk score to auto-escalate priority, cutting manual triage time by 50%.
- Delivered live dashboards (Recharts) with real-time stats, a zone-by-zone water-quality monitor against WHO thresholds, neighbourhood alerts, an emergency incident board, and an admin audit log.
- github.com/avinash09-cmd/NEST

Certifications

[Google IT Support Professional Certificate](#) — Google Career Certificates (2026)

[Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate](#) — Oracle (2025)

[The Bits and Bytes of Computer Networking](#) — Google / Coursera

Achievements

Top-5 Finalist (out of 200+ teams), Anti-Piracy Hackathon — WAVES Summit for MSME Industry, Ministry of Maharashtra (2026)

National-Level multi-sport athlete — Basketball, Swimming

Positions of Responsibility

2025 – **Vice President**, *Metaversity Club*, *VIT Bhopal University*

Present Led a team of 23 members, coordinating events and technical initiatives for the university's metaverse and emerging-tech community.